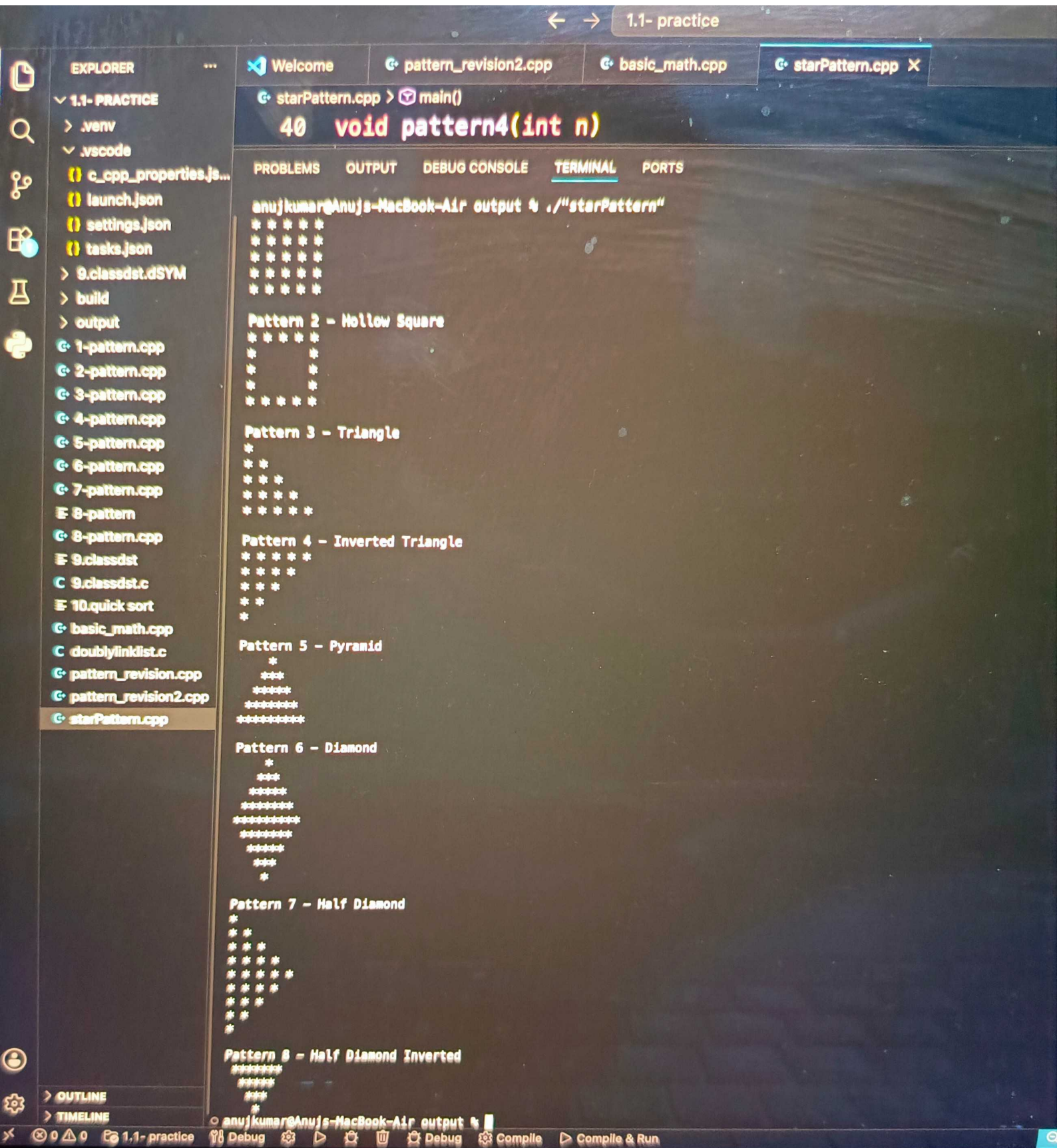




EXPLORER

1.1- PRACTICE

- .env
- .vscode
 - c_cpp_properties.js...
 - launch.json
 - settings.json
 - tasks.json
- 9.classdst.dSYM
- build
- output
- 1-pattern.cpp
- 2-pattern.cpp
- 3-pattern.cpp
- 4-pattern.cpp
- 5-pattern.cpp
- 6-pattern.cpp
- 7-pattern.cpp
- 8-pattern
- 8-pattern.cpp
- 9.classdst
- 9.classdst.c
- 10.quick sort
- basic_math.cpp
- doublylinklist.c
- numberPatterns.cpp
- pattern_revision.cpp
- pattern_revision2.cpp
- starPatterns.cpp

OUTLINE

TIMELINE

Welcome pattern_revision2.cpp basic_math.cpp starPatterns.cpp

numberPatterns.cpp > pattern6(int)

```

85 void pattern5(int n)
102     for(int i=mid-1;i>=1;i--)
105         cout<<" ";
106
107     for(int j=1;j<=n-2*(i-1);j++)
108         cout<<i+2<<" ";
109
110     cout<<endl;
  
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

anujkumar@Anuj's-MacBook-Air output % ./"numberPatterns"

Pattern 1 - Number Square

```

1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
  
```

Pattern 2 - Incrementing Pattern

```

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
  
```

Pattern 3 - Right Triangle

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
  
```

Pattern 4 - Inverted Diamond

```

3 3 3 3 3
4 4 4 4
5 5 5
6 6
7
  
```

Pattern 5 - Sandwich Pattern

```

3 3 3 3 3
4 4 4
5
4 4 4
3 3 3 3 3
  
```

Pattern 6 - Incrementing Diamond

```

3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
  
```

anujkumar@Anuj's-MacBook-Air output %

① Star patterns

① Square

* * *

* * *

* * *

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        cout << " ";
    }
}
```

cout << endl;

}

② Hollow square

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

```
for (int i = 1; i <= n; i++) {
```

```
    for (int j = 1; j <= n; j++) {
```

```
        if (i == 1 || i == n ||
            j == 1 || j == n) {
```

```
            cout << " ";
```

```
        } else
```

```
            cout << " ";
```

```
    }
```

```
    cout << endl;
```

```
}
```

③ Triangle

*

* *

* * *

* * * *

```
for (int i = 1; i <= n; i++) {
```

```
    for (int j = 1; j <= i; j++) {
```

```
        cout << " ";
```

```
    }
```

```
    cout << endl;
```

④ Inverted triangle

```
for(int i=n; i>=1; i--){
    for(int j=1; j<=i; j++){
        cout << "x ";
    }
    cout << endl;
}
```

```
* * * *
* * *
* *
*
```

⑤ Pyramid

```
row → for(int i=1; i<=n; i++){
    column {
        for(int j=1; j<=n-i; j++){
            cout << " "; // (space)
        }
        for(int j=1; j<=2*i-1; j++){
            cout << "x ";
        }
        cout << endl;
    }
}
```

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *
```

⑥ ~~Half~~ Diamond

// first half

```
row → for(int i=1; i<=n; i++){
    column {
        for(int j=1; j<=n-i; j++){
            cout << " ";
        }
        for(int j=1; j<=2*i-1; j++){
            cout << "x ";
        }
        cout << endl;
    }
}
```

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *
```

// second half

```

row for (int i = n-1; i >= 1; i--) {
    for (int j = 1; j <= n-i; j++) {
        cout << " ";
    }
    for (int j = 1; j <= 2*i-1; j++) {
        cout << "x";
    }
    cout << endl;
}

```

Column

⑦ Half Diamond.

// first half

```

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        cout << "x";
    }
}

```

cout << endl;

}

Second
half

```

for (int i = n-1; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        cout << "x";
    }
}

```

cout << endl;

}

⑧ Half Diamond Inverted.

```

for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        cout << " ";
    }
}

```

(Space)

```

for (int j=1; j<2*(n-i); j++) {
    cout<<"*";
}
cout<<endl;
}

```

②

Number patterns:-

① Square

Similar as 1 square

change (cout<<"*";)

cout<<"1";

1 1 1

1 1 1

1 1 1

②

Incrementing pattern

1 2 3 4

1 2 3 4

```

for (int i=1; i<=n; i++)

```

1 2 3 4

```

{ for (int j=1; j<=n; j++) {

```

1 2 3 4

```

    cout<<j;

```

```

}

```

```

    cout<<endl;

```

```

}

```

③

Right triangle Numbers

1

1 2

```

for (int i=1; i<=n; i++) {

```

1 2 3

```

    for (int j=1; j<=i; j++) {

```

1 2 3 4

```

        cout<<j;

```

```

    }
    cout<<endl;
}

```

④ Inverted Diamond

```

for(int i=3; i<=n+3; i++)
for(int j=n+2; j>=3; j--){
    for(int k=
for(int l=n; l>=1; l--){
    for(int m=1; m<=j; m++){
        cout<< l+2;
    }
    cout<<endl;
}
}

```

33333
4444
555
66
7

```

for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++){
        cout<< j+2;
    }
    cout<<endl;
}

```

⑤ Sandwich pattern .

```

for(int i=0; i<n; i++){
    for(int j=0; j<=i; j++){
        cout<<" ";
    }
}

```

33333

444

5

444

33333

first
half

```

for(int j=1; j<=2*(n-i); j++){
    cout<< i+2;
}
cout<<endl;
}

```

Second half:

```

for(int i=n-1; i>=1; i--){
    for(int j=1; j<=2*(i-1); j++){
        cout<<" ";
    }
    for(int j=1; j<=2*(i-1); j++){
        cout<< i+2<<" ";
    }
    cout<<endl;
}
}

```

Incrementing Diamond pattern

3
4 4
5 5 5
6 6 6 6
5 5 5
4 4
3

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= i; j++)  
    {  
        cout << i + 2 << " ";  
    }  
    cout << endl;  
}
```

```
// lower (second half)  
for (int i = n - 1; i >= 1; i--) {  
    for (int j = 1; j <= i; j++)  
    {  
        cout << i + 2 << " ";  
    }  
    cout << endl;  
}
```